Chapter 9 Review

Campbell Biology, AP Edition, 12e

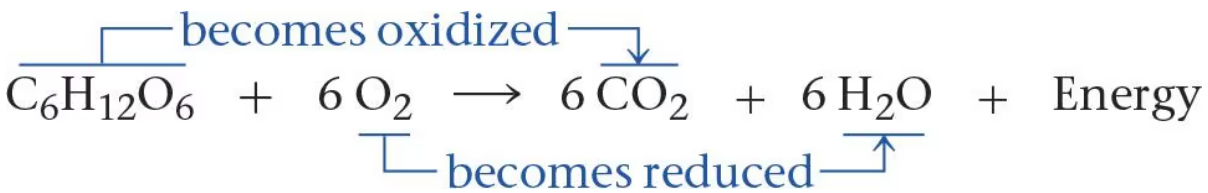
Chapter 9 Review

Summary of Key Concepts

What controls the different mechanisms and structural features that organisms use to capture, store, and use energy? (**BIG IDEAS 2, 3, & 4)**

Concept 9.1: Catabolic pathways yield energy by oxidizing organic fuels

- Cells break down glucose and other organic fuels to yield chemical energy in the form of ATP. **Fermentation** is a process that results in the partial degradation of glucose without the use of oxygen O2. The process of **cellular respiration** is a more complete breakdown of glucose. In **aerobic respiration**, O2 is used as a reactant; in *anaerobic respiration*, other substances are used in place of O2.
- The cell taps the energy stored in food molecules through **redox reactions**, in which one substance partially or totally shifts electrons to another. **Oxidation** is the total or partial loss of electrons, while **reduction** is the total or partial addition of electrons. During aerobic respiration, glucose C6H12O6 is oxidized to CO2, and O2 is reduced to H2O:



- Electrons lose potential energy during their transfer from glucose or other organic compounds to O2. Electrons are usually passed first to **NAD+**, reducing it to **NADH**, and are then passed from NADH to an **electron transport chain**, which conducts the electrons to O2 in energy-releasing steps. The energy that is released is used to make ATP.
- Aerobic respiration occurs in three stages: (1) **glycolysis**, (2) pyruvate oxidation and the **citric acid cycle**, and (3) **oxidative phosphorylation** (electron transport and chemiosmosis).

Describe the difference between the two processes in cellular respiration that produce ATP: oxidative phosphorylation and substrate-level phosphorylation.

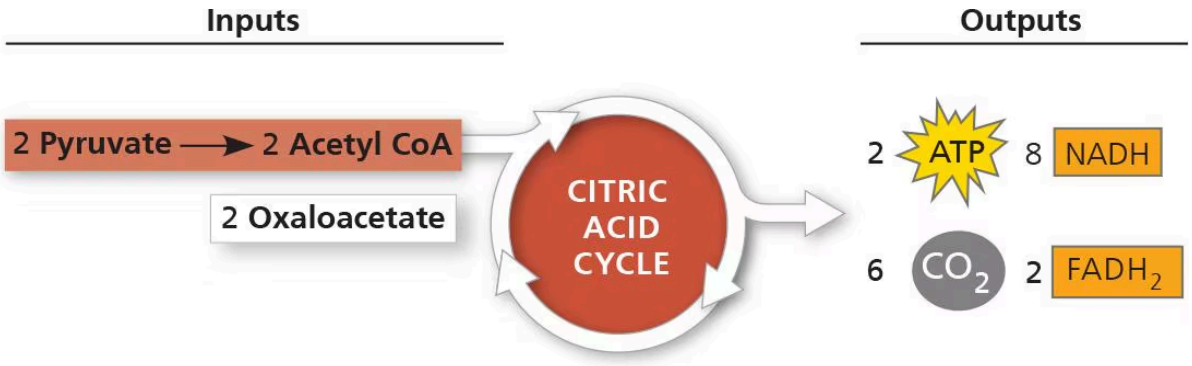
Concept 9.2: Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

- Glycolysis ("splitting of sugar") is a series of reactions that breaks down glucose into two pyruvate molecules, which may go on to enter the citric acid cycle, and nets 2 ATP and 2 NADH per glucose molecule.

Which reactions in glycolysis are the source of energy for the formation of ATP and NADH?

Concept 9.3: After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules

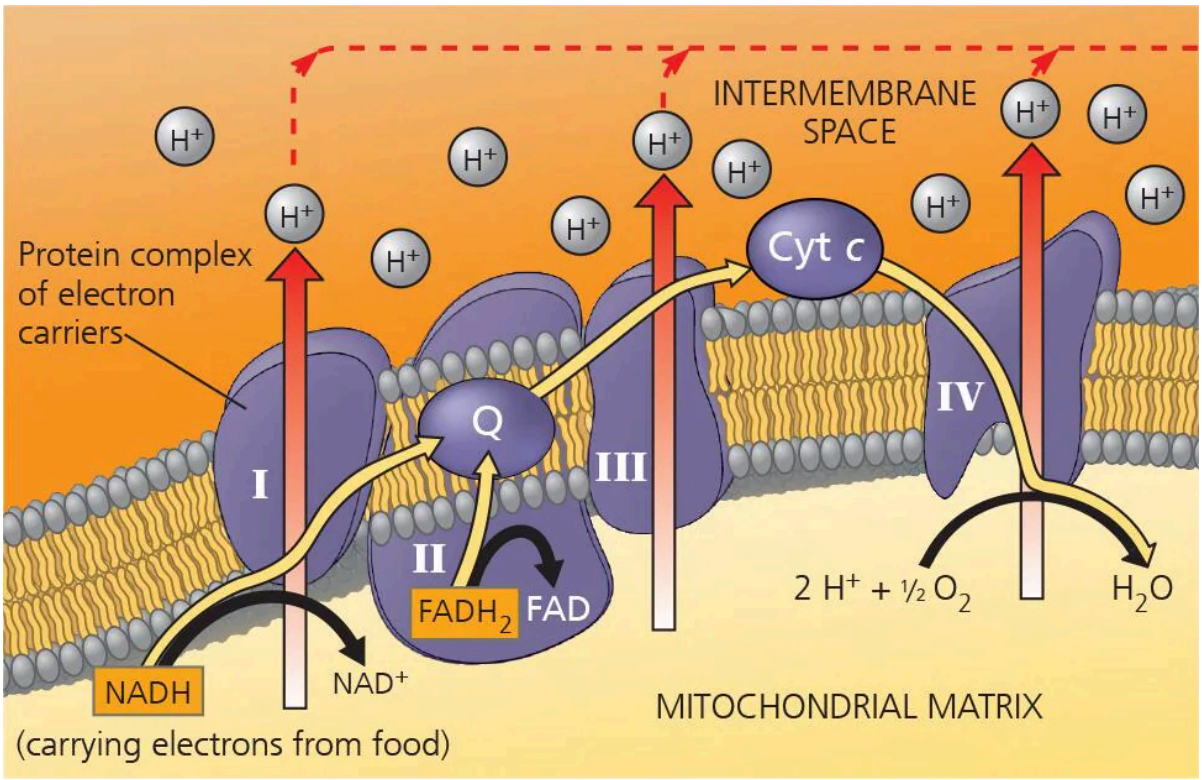
- In eukaryotic cells, pyruvate enters the mitochondrion and is oxidized to **acetyl CoA**, which is further oxidized in the citric acid cycle.



What molecular products indicate the complete oxidation of glucose during cellular respiration?

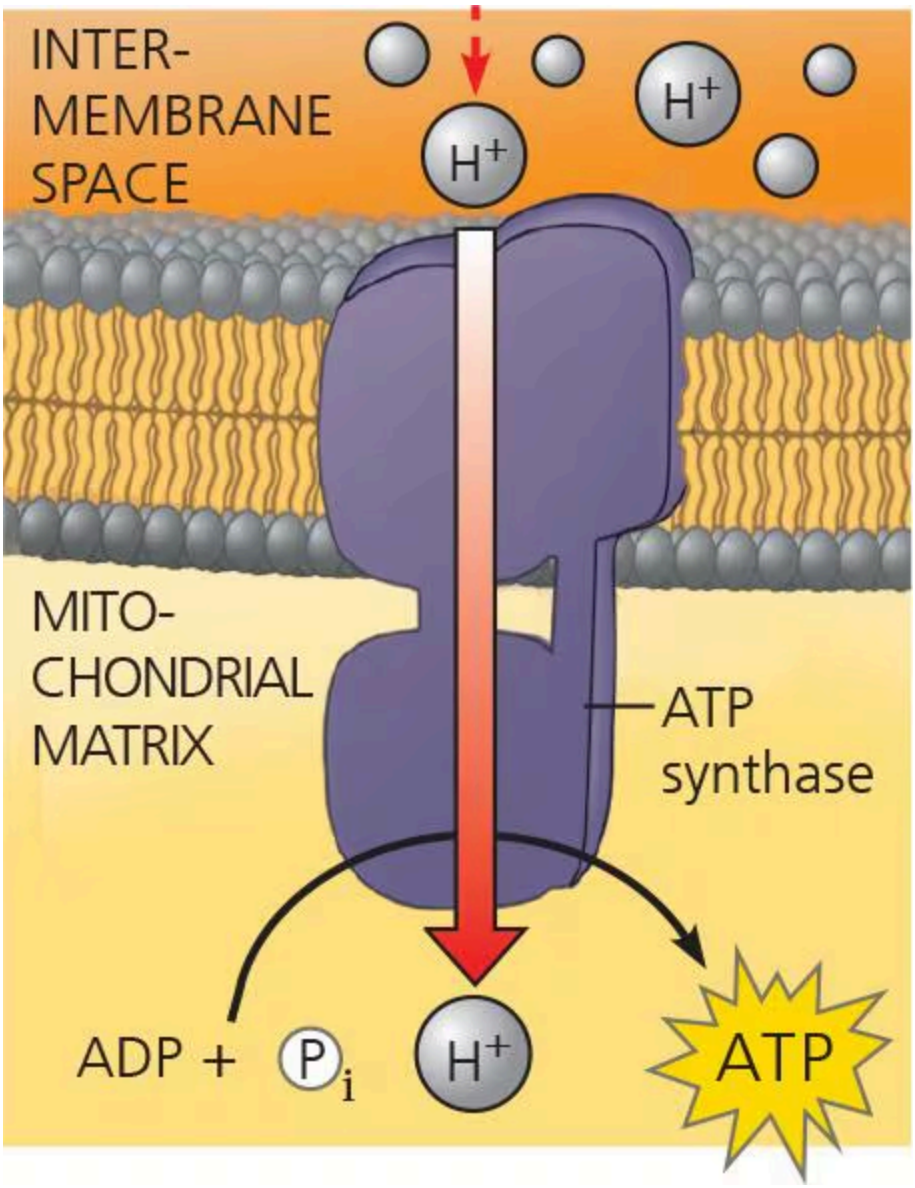
Concept 9.4: During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

- NADH and FADH2 transfer electrons to the electron transport chain. Electrons move down the chain, losing energy in several energy-releasing steps. Finally, electrons are passed to O2, reducing it to H2O.



In the electron transport chain, electron transport through a series of protons embedded in the inner mitochondrial membrane causes protons to be pumped, creating an H-plus gradient across the membrane. The inner mitochondrial membrane separates mitochondrial matrix from the intermembrane space. N A D H carries electrons from food to the electron transport chain, a series of protein complexes of electron carriers embedded in the inner mitochondrial membrane. The electrons are passed through Complex 1, and a proton is pumped into the intermembrane space. The electrons then pass downhill to Q, then to Complex 3, where a proton is pumped into the intermembrane space. The electrons pass through Cyt c and then to Complex 4, where a proton is pumped into the intermembrane space, and at the bottom of this last complex, electrons join 2 H plus and one-half O 2 to form H 2 O. Electrons from F A D H 2 can also join the electron transport chain at Q, from which point they will follow the same path as electrons from N A D H.

- Along the electron transport chain, electron transfer causes protein complexes to move H+ from the mitochondrial matrix (in eukaryotes) to the intermembrane space, storing energy as a **proton-motive force** (H+ gradient). As H+ diffuses back into the matrix through **ATP synthase**, its passage drives the phosphorylation of ADP to form ATP, called **chemiosmosis**.



Protons pumped into the intermembrane space during the electron transport chain fall back down their concentration gradient through the A T P synthase (embedded in the inner mitochondrial membrane), causing inorganic phosphate to be added to A D P, forming A T P.

- About 34% of the energy stored in a glucose molecule is transferred to ATP during cellular respiration, producing a maximum of about 32 ATP.

Briefly explain the mechanism by which ATP synthase produces ATP. List three locations in which ATP synthases are found.

Concept 9.5: Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen

- Glycolysis nets 2 ATP by substrate-level phosphorylation, whether O_2 is present or not. Under anaerobic conditions, anaerobic respiration or fermentation can take place. In anaerobic respiration, an electron transport chain is present with a final electron acceptor other than oxygen. In fermentation, the electrons from NADH are passed to pyruvate or a derivative of pyruvate, regenerating the NAD^+ required to oxidize more glucose. Two common types of fermentation are **alcohol fermentation** and **lactic acid fermentation**.
- Fermentation, anaerobic respiration, and aerobic respiration all use glycolysis to oxidize glucose, but they differ in their final electron acceptor and whether an electron transport chain is used (respiration) or not (fermentation). Respiration yields more ATP; aerobic respiration, with O_2 as the final electron acceptor, yields about 16 times as much ATP as does fermentation.
- Glycolysis occurs in nearly all organisms and is thought to have evolved in ancient prokaryotes before there was O_2 in the atmosphere.

Which process yields more ATP, fermentation or anaerobic respiration? Explain.

Concept 9.6: Glycolysis and the citric acid cycle connect to many other metabolic pathways

- Catabolic pathways funnel electrons from many kinds of organic molecules into cellular respiration. Many carbohydrates can enter glycolysis, most often after conversion to glucose. Amino acids of proteins must be deaminated before being oxidized. The fatty acids of fats undergo **beta oxidation** to two-carbon fragments and then enter the citric acid cycle as acetyl CoA. Anabolic pathways can use small molecules from food directly or build other substances using intermediates of glycolysis or the citric acid cycle.
- Cellular respiration is controlled by allosteric enzymes at key points in glycolysis and the citric acid cycle.

Describe how the catabolic pathways of glycolysis and the citric acid cycle intersect with anabolic pathways in the metabolism of a cell.